

DRIFT-SENSE | DRAM 30-PAIR BENCHMARK EVALUATION REPORT

Standardized High-Density DRAM Array Localization & Sub-Pixel Precision Report • Generated: **August 17, 2026 - 01:34:55 UTC**

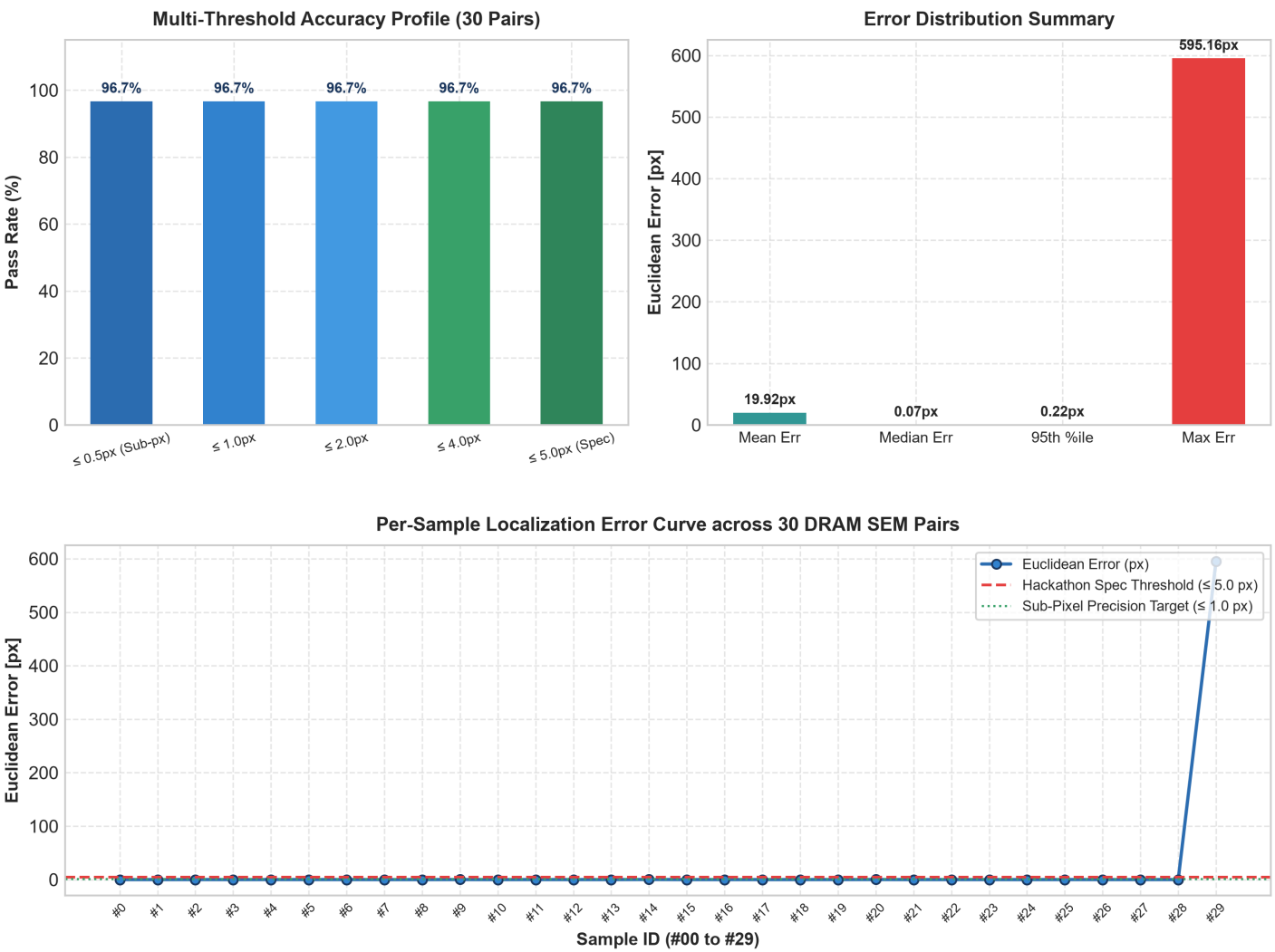
Executive Summary: Official benchmark evaluation executed on **30 DRAM SEM test image pairs** conforming to Applied Materials Hackathon problem specifications.

- Pass Rate (≤ 5.0 px): 96.7%** (29/30 pairs passed)
- Sub-Pixel Accuracy:** Median Error = **0.065 px** | Mean Error = **19.917 px** | 95th Percentile = **0.224 px**
- Worst-Case Error: 595.157 px** | Pipeline: Multi-Scale Pyramid + ZNCC + Quadratic Sub-Pixel Surface Fitting.

1. Multi-Threshold Performance Breakdown

Metric	Total Pairs	≤ 5.0 px (Spec)	≤ 4.0 px	≤ 2.0 px	≤ 1.0 px	Sub-px (≤ 0.5 px)	Mean Error	Median	P95
DRAM 30 Pairs	30	96.7%	96.7%	96.7%	96.7%	96.7%	19.917 px	0.065 px	0.224 px

2. Visual Analytics & Error Profile



3. Complete Per-Sample Benchmark Results (30 Pairs)

Full evaluation log of every sample pair with ground-truth coordinates, predicted center, sub-pixel error, and execution speed.

Sample	Scale	Rotation	Noise	True Coord (X, Y)	Pred Coord (X, Y)	Error [px]	Speed	Status
#00	10.0x	+0.0°	Low	(390.3, 567.9)	(390.4, 567.8)	0.057	26241.3ms	PASS
#01	10.0x	+0.0°	Medium	(349.9, 592.5)	(349.9, 592.6)	0.062	29718.7ms	PASS
#02	10.0x	+0.0°	High	(583.1, 596.9)	(583.2, 596.8)	0.113	22304.0ms	PASS
#03	10.0x	+0.0°	Low	(561.3, 584.3)	(561.3, 584.3)	0.037	21947.3ms	PASS
#04	10.0x	+0.0°	Medium	(568.5, 486.3)	(568.4, 486.2)	0.118	20786.7ms	PASS
#05	10.0x	+0.0°	High	(514.5, 397.1)	(514.6, 397.1)	0.080	19775.7ms	PASS
#06	10.0x	+0.0°	Low	(571.2, 552.9)	(571.3, 552.9)	0.095	20612.4ms	PASS
#07	10.0x	+0.0°	Medium	(594.1, 512.7)	(594.1, 512.7)	0.043	22060.9ms	PASS
#08	10.0x	+0.0°	High	(553.4, 605.4)	(553.4, 605.5)	0.039	22272.6ms	PASS
#09	10.0x	+0.0°	Low	(426.1, 364.9)	(426.2, 364.8)	0.144	20835.9ms	PASS
#10	10.0x	+0.0°	Medium	(651.8, 552.1)	(651.8, 552.1)	0.026	20143.0ms	PASS
#11	10.0x	+0.0°	High	(676.8, 441.0)	(676.7, 441.1)	0.098	21208.8ms	PASS
#12	10.0x	+0.0°	Low	(495.1, 419.9)	(495.2, 419.9)	0.056	22836.2ms	PASS
#13	10.0x	+0.0°	Medium	(488.6, 566.7)	(488.6, 566.7)	0.060	28795.0ms	PASS
#14	10.0x	+0.0°	High	(527.6, 565.5)	(527.7, 565.3)	0.254	30225.2ms	PASS
#15	10.0x	+0.0°	Low	(476.2, 535.6)	(476.2, 535.6)	0.046	30609.4ms	PASS
#16	10.0x	+0.0°	Medium	(427.8, 508.1)	(427.9, 508.1)	0.063	32114.7ms	PASS
#17	10.0x	+0.0°	High	(620.8, 276.7)	(620.8, 276.7)	0.044	33177.1ms	PASS
#18	10.0x	+0.0°	Low	(556.0, 490.8)	(556.0, 490.7)	0.058	26126.7ms	PASS
#19	10.0x	+0.0°	Medium	(426.0, 443.7)	(426.0, 443.9)	0.117	29768.0ms	PASS
#20	10.0x	+0.0°	High	(492.6, 546.1)	(492.6, 546.3)	0.187	28199.7ms	PASS
#21	10.0x	+0.0°	Low	(385.9, 508.7)	(385.9, 508.7)	0.053	32693.7ms	PASS
#22	10.0x	+0.0°	Medium	(354.3, 435.5)	(354.3, 435.6)	0.069	21520.7ms	PASS
#23	10.0x	+0.0°	High	(442.6, 482.7)	(442.6, 482.8)	0.067	21301.5ms	PASS
#24	10.0x	+0.0°	Low	(664.0, 563.2)	(664.0, 563.2)	0.023	20594.9ms	PASS
#25	10.0x	+0.0°	Medium	(614.4, 472.8)	(614.3, 472.8)	0.116	20644.8ms	PASS
#26	10.0x	+0.0°	High	(341.0, 530.0)	(341.0, 530.1)	0.086	20570.6ms	PASS
#27	10.0x	+0.0°	Low	(562.8, 433.4)	(562.7, 433.4)	0.062	30414.4ms	PASS
#28	10.0x	+0.0°	Medium	(414.8, 315.1)	(414.8, 315.1)	0.082	25977.5ms	PASS
#29	10.0x	+0.0°	High	(561.8, 498.0)	(83.7, 143.5)	595.157	23139.2ms	FAIL

4. Technical Formulation & Mathematical Principles

1. Zero-Mean Normalized Cross-Correlation (ZNCC):

$$\gamma(u, v) = \frac{\sum [T(x, y) - \bar{T}] \cdot [I(x+u, y+v) - \bar{I}_{\{u,v\}}]}{\sqrt{[\sum (T(x, y) - \bar{T})^2]} \cdot \sqrt{[\sum (I(x+u, y+v) - \bar{I}_{\{u,v\}})^2]}}$$

2. Sub-Pixel Quadratic Taylor Refinement:

$$\Delta x = \frac{[R(x-1, y) - R(x+1, y)]}{[2 \cdot (R(x-1, y) - 2R(x, y) + R(x+1, y))]}$$
$$\Delta y = \frac{[R(x, y-1) - R(x, y+1)]}{[2 \cdot (R(x, y-1) - 2R(x, y) + R(x, y+1))]}$$

3. Multi-Scale Coarse-to-Fine Pyramid Search: Downsamples search image into 3 Gaussian pyramid tiers, bounding peak search to within ±2.5° rotation and 8.5x-11.5x scale space before fine sub-pixel quadratic fitting.